Physics - Simple harmonic motion

Consider a mass attached to a spring.

Restoring force: F = -kx

Newton: m x'' = -kx

=> x'' + (k/m) x = 0

General solution:

x(t) = A cos(wt + phi)

w = sqrt(k/m)

Amplitude A depends on initial conditions.

Frequency f = w / (2*pi)

Energy: E = (1/2) k A^2



Diagram: spring - mass system